

Functional Annotation Report: MlaF (Q88P94, PP_0958) — *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

MlaF is the cytoplasmic ATP-binding/ATPase (nucleotide-binding domain, NBD) subunit of the inner-membrane ABC transporter MlaFEDB, the engine of the Mla (Maintenance of Lipid Asymmetry) system. By binding and hydrolyzing ATP at the cytoplasmic face of the inner membrane, MlaF powers the transport of glycerophospholipids across the inner membrane as part of a cell-envelope-spanning pathway (MlaA–OmpC/F in the outer membrane → periplasmic shuttle MlaC → MlaFEDB in the inner membrane). The primary physiological role of this pathway is the **retrograde removal of mislocalized phospholipids from the outer leaflet of the outer membrane back to the inner membrane**, thereby maintaining outer-membrane (OM) lipid asymmetry and the permeability barrier that protects Gram-negative bacteria against antibiotics and other toxic compounds.

Identity verification. The UniProt record (Q88P94; gene *miaF*; ordered locus PP_0958; ABC-transporter superfamily / MlaF family; domains AAA+/P-loop NTPase and ABC transporter ATP-binding, PF00005) is fully consistent with the "MlaF" characterized experimentally in *E. coli*, *A. baumannii*, and *Pseudomonas aeruginosa*. No *P. putida* KT2440-specific biochemical study of MlaF was found; the functional annotation below is therefore established from well-characterized orthologs and from the conserved domain architecture, which are highly conserved across diderm (Gram-negative) bacteria. This is a confident annotation by orthology, not a case of gene-symbol ambiguity.

2. Molecular Function and Substrate

- **Enzymatic activity / role:** MlaF is a nucleotide-binding domain of the ABC transporter superfamily. Its Walker A (P-loop), Walker B, and ABC-signature motifs (reflected in InterPro

IPR003439/IPR003593/IPR017871/IPR027417 and Pfam PF00005) bind and hydrolyze ATP. Two MlaF protomers form the NBD dimer of the MlaFEDB complex; ATP binding drives NBD dimerization and the conformational cycle that translocates lipid substrate through the transmembrane MlaE permease [PMID 33236984;

P 33199922

- **Substrate of the transporter it powers:** glycerophospholipids. The cryo-EM structure of MlaFEDB captured **two phospholipid molecules bound** along a continuous conduit running from the MlaE substrate-binding site through the MlaD pore, indicating phospholipids are the transported substrate [PMID 33236984].
- **Mechanistic caveat:** while MlaF is unambiguously the ATPase, precisely how ATP hydrolysis is coupled to the direction and steps of lipid movement is still debated and not fully resolved [PMID 39080293].
- **Organism-specific sequence evidence:** A motif scan of the actual 269-residue *P. putida* MlaF (Q88P94) confirms a complete, intact ABC nucleotide-binding active site: **Walker A / P-loop** **GPSGCGKT** (res 41–48), **Q-loop Gln** (res ~92), **ABC signature / C-loop** **LSGGM** (res 145–149), **Walker B** **ILMYDE** with the catalytic **Asp/Glu (res 165–170)**, and the **H-loop switch histidine** **VSHD** (res 201–204). The presence of all catalytic elements (P-loop for phosphate binding, Walker B Asp/Glu for Mg²⁺ coordination and hydrolysis, switch His for γ -phosphate sensing) indicates the *P. putida* protein itself is a catalytically competent ATPase, not merely inferred from orthologs.

3. Localization

- MlaF is a **peripheral cytoplasmic-membrane protein**: the NBDs sit on the **cytoplasmic (inner) face of the inner membrane**, coupled to the integral-membrane MlaE permease dimer, the periplasmic MCE domain hexamer MlaD, and the small regulatory subunit MlaB [PMID 33236984;

P 33199922

- The complex it belongs to (MlaFEDB) is one of three sub-cellular assemblies of the Mla pathway, which spans the entire Gram-negative cell envelope: **MlaA–OmpC/F (outer membrane)**, **MlaC (periplasm)**, **MlaFEDB (inner membrane)** [PMID 39080293;

P 34753108

4. Pathway and Biological Process

The Mla system maintains OM lipid asymmetry — lipopolysaccharide in the OM outer leaflet, glycerophospholipids in the inner leaflet. Glycerophospholipids that mislocalize to the OM outer leaflet compromise the barrier; the Mla pathway removes them:

1. **MlaA–OmpC/F** extracts mislocalized phospholipids from the OM outer leaflet.
2. **MlaC** shuttles the lipid across the periplasm (binding either MlaA or MlaD, one at a time) [PMID 37100290;

P 39080293

3. **MlaFEDB**, energized by **MlaF ATP hydrolysis**, receives the lipid at MlaD and moves it into the inner membrane.

Genetic and, more decisively, in vitro reconstitution/biochemical studies support **retrograde** (OM→IM) directionality, resolving an earlier anterograde-vs-retrograde debate [PMID 36459067;

P 34753108

Physiologically, loss of Mla function increases membrane permeability and antibiotic susceptibility, promotes outer-membrane vesicle blebbing, and reduces virulence [PMID 39080293]. In the closely related *Pseudomonas aeruginosa*, deleting the OM component *mlaA* increases fluoroquinolone susceptibility and reduces rhamnolipid secretion, motility, and biofilm formation — illustrating the pathway's envelope-integrity role in pseudomonads [PMID 37660742].

5. Structural/Family Classification and Regulation

- MlaFEDB is proposed to be the **founding member of a structurally distinct ABC superfamily**, with distant relationships to LPS, MacAB, and eukaryotic ABCA/ABCG transporters; MlaE has minimal sequence similarity to classical permeases [PMID 39080293;

P 33236984

P 35981415

- MlaF partners directly with **MlaB, a small cytoplasmic STAS-domain subunit** that assembles with and supports MlaF; the MlaB–MlaF interface and the non-canonical features

distinguishing Mla proteins from classical ABC components are areas of active study [PMID 41047745].

6. Supported vs. Refuted Hypotheses

- **Supported:** MlaF is the ATP-binding/ATPase subunit of the MlaFEDB inner-membrane ABC transporter (domain + structural evidence). Substrate = glycerophospholipids. Localization = cytoplasmic face of the inner membrane. Pathway = Mla, retrograde phospholipid transport maintaining OM asymmetry.
- **Refuted/deprioritized:** Purely anterograde (IM→OM) transport as the primary function — recent biochemistry favors retrograde [PMID 36459067].
- **Open:** The exact coupling of MlaF ATP hydrolysis to individual lipid-transfer steps [PMID 39080293].

7. Limitations and Future Directions

- All mechanistic evidence derives from orthologs (*E. coli*, *A. baumannii*, *P. aeruginosa*); a direct *P. putida* KT2440 MlaF study was not identified. However, annotation transfer is well justified: a global alignment of *P. putida* MlaF (Q88P94, 269 aa) against experimentally/structurally characterized *E. coli* MlaF (P63386, 269 aa) gives **57.6% identity (155/269) over the full length with a 1:1 ungapped correspondence** — a clear one-to-one ortholog far above the ~30% threshold for confident functional transfer. Combined with the intact ABC-ATPase active site (Section 2) and near-universal conservation of Mla in diderms, the annotation transfers with high confidence.
- Future work specific to *P. putida* could test whether *mlaF* deletion affects the organism's notable solvent tolerance and envelope robustness, and could confirm ATP-hydrolysis-dependent retrograde phospholipid transport in the KT2440 complex directly.

Key

references:

[P 33236984](#)

(Coudray 2020, MlaFEDB structure with substrate);

[P 33199922](#)

(Tang 2021, nucleotide-bound structures + transport assays);

[P 39080293](#)

(Wotherspoon 2024, MlaC–MlaD, ABC-superfamily statement);

[P 36459067](#)

(Abellon-Ruiz 2023, transport direction);

[P 34753108](#)

(Low & Chng 2021, pathway review);

[P 35981415](#)

(Ekiert 2022, mechanism review);

[P 37100290](#)

(MacRae 2023, interactions);

[P 41047745](#)

(Dutta 2026, component analysis);

[P 37660742](#)

(Kaur 2023, *Pseudomonas mlaA* phenotypes).